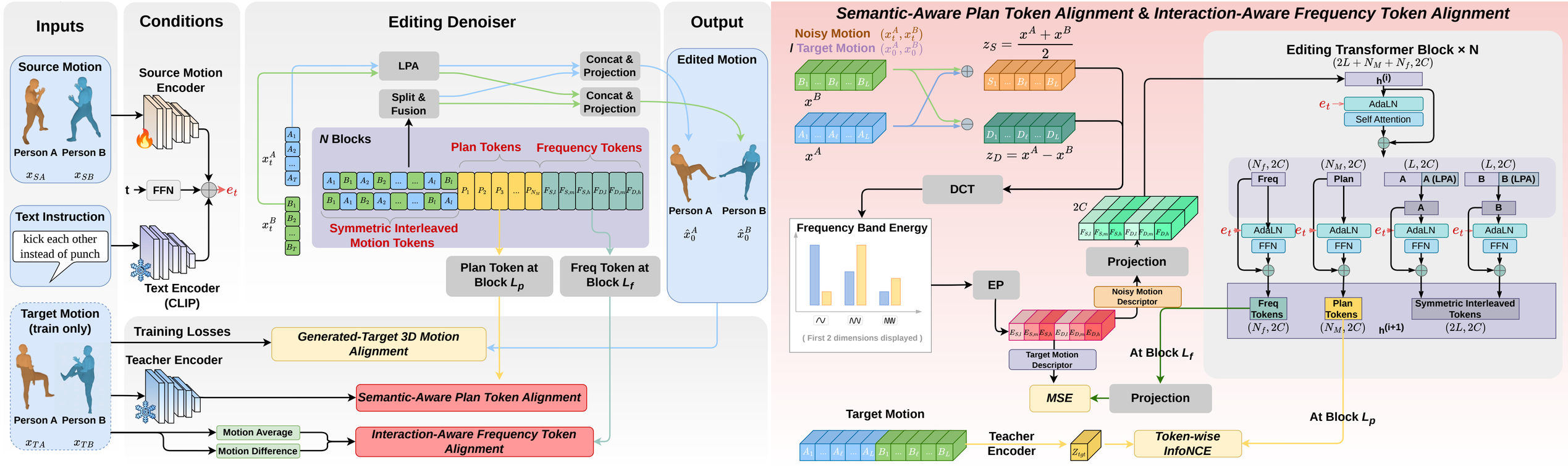
Inputs
Conditions
Editing Denoiser
Output
Source Motion
Person A
Person B
Source Motion Encoder
t
FFN
Text Instruction
kick each other instead of punch
Text Encoder (CLIP)
Target Motion (train only)
Teacher Encoder
Training Losses
Generated-Target 3D Motion Alignment
Semantic-Aware Plan Token Alignment
Motion Average
Motion Difference
Interaction-Aware Frequency Token Alignment
LPA
Split & Fusion
Concat & Projection
N Blocks
Plan Tokens
Frequency Tokens
Symmetric Interleaved Motion Tokens
Plan Token at Block L_p
Freq Token at Block L_f
Edited Motion
Semantic-Aware Plan Token Alignment & Interaction-Aware Frequency Token Alignment
Noisy Motion (x_t^A, x_t^B)
/ Target Motion (x_0^A, x_0^B)
$z_S = \frac{x^A + x^B}{2}$
$z_D = x^A - x^B$
DCT
Frequency Band Energy
(First 2 dimensions displayed)
EP
Projection
Noisy Motion Descriptor
Target Motion Descriptor
MSE
At Block L_f
Editing Transformer Block × N
$(2L + N_M + N_f, 2C)$
AdaLN
Self Attention
Freq
Plan
A
A (LPA)
B
B (LPA)
FFN
Freq Tokens
Plan Tokens
Symmetric Interleaved Tokens
At Block L_p
Target Motion
Teacher Encoder
Token-wise InfoNCE